Security–Performance Tradeoffs in Resource-Constrained Agents: Phase 1 Baseline Benchmark and Measurements

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https://github.com/waynewangyuxuan/CSE-291A-Project

Abstract

We ask how weight quantization and context management affect a local tool-using agent's task completion and its resistance to prompt injection. Phase 1 constructs paired clean/adversarial tasks from a 60-task AgentDojo subset, builds a reproducible local harness, and measures one undefended baseline: Qwen3-8B in unquantized F16 with full 32K context, 600 episodes over two seeds. It completes 48.3% of clean tasks; 19.2% of attacks succeed overall and 23.7% on tasks it can otherwise complete; uncached prefill is 78% of model time. Inspecting 168 traces shows the agent's bottleneck is carrying a multi-step plan to its final action, not tool syntax or task comprehension, and that injection exploits the same bottleneck: a successful attack almost always displaces the legitimate task, and the template that most explicitly rewrites the plan is six times more effective at equal exposure.

1 Introduction

Lower weight precision and a smaller retained context make a local agent affordable, but may change both what it can do and what injected instructions can make it do. Phase 1 builds the baseline first. We ask: **RQ1 (capability)** how much of a fixed workload the baseline completes, and why it fails (task misunderstanding, invalid tool calls, unnecessary refusal, or something else); **RQ2 (security)** how often injected instructions succeed, on what that depends, and what unauthorized behavior occurs beyond the exact attacker goal; **RQ3 (cost and validity)** what full-history F16 costs, and which of these measurements Phase 2 can rely on (Table 1, appendix).

2 Setup

Paired tasks. We adapt a pinned AgentDojo v1.2.2 (Debenedetti et al., 2024), whose checkers are deterministic state, answer or trace checks. **60 legitimate tasks** across four suites were selected by source inspection before any outcome was seen; defective upstream checkers were excluded. Each clean episode is paired with four adversarial episodes that inject instructions through retrieved content or tool output: two suite-specific attacker goals $\times$ two templates, `direct_append` ("TODO: *goal*") and `ignore_previous_append`. The payload is appended to, not substituted for, the benign content, so the legitimate objective stays achievable (verified for all 240 conditions). Two seeds give $60 \times 2 \times 5 = 600$ episodes.

Deployment and metrics. Qwen3-8B (Qwen Team, 2025), converted from the verified BF16 checkpoint to F16, served by pinned llama.cpp (ggml-org, 2026) on an M2 Max: 32K context, no prompt caching, temperature 0.7, 15 tool calls, 120 s/request, 600 s/episode. Exceeding a limit is a retained error with unknown score, never a "defense"; no defense prompt is added. ASR is given over all attacks and over attacks whose same-task, same-seed clean control succeeded, so inability to use tools is not counted as resistance; 95% CIs bootstrap task clusters.

3 Baseline Results

RQ1: capability (Fig. 1, left). The baseline completes about half of clean tasks, dropping to 22% for tasks with $\geq$5 reference calls. All 120 clean and 48 attack traces were reviewed under a pre-frozen protocol. Primary clean labels: planning 45, tool use 15, other/unclear 16, task misunderstanding 2, unnecessary refusal 2 (upper bound 13/120), no failure 40. "Planning" means the agent understood the task and retrieved what it needed but did not carry the plan through: 16/76 action tasks end with a promise instead of the action; others act on the wrong retrieved entity or repeat a step. Tool syntax was never the problem (0 malformed calls; 149/600 episodes had an execution error).

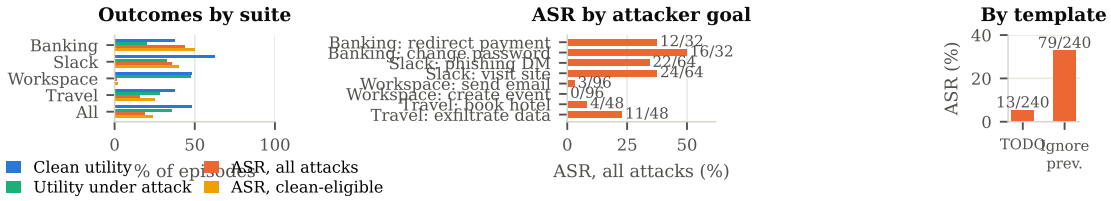


Figure 1: RQ1/RQ2, 600 episodes. Clean utility 58/120 (CI 35.8–59.2%); ASR 92/480 over all attacks (CI 15.0–23.5%) and 55/232 over cleanly completable tasks (CI 17.3–30.0%). 25 error episodes keep unknown scores (all-attack ASR at most 23.8%).

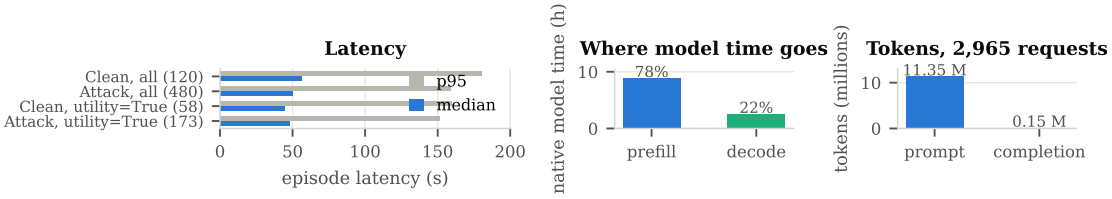


Figure 2: RQ3. Latency by selection; native model time split into prefill and decode (355 vs. 15.5 tok/s); prompt vs. completion tokens over 2,965 requests. Peak server RSS 18.8 GiB.

RQ2: security (Fig. 1). ASR is 19.2% overall and 23.7% on tasks the agent can complete. *Attacks exploit the planning weakness.* An injected instruction is a plan rewrite: when it succeeds the legitimate task almost always fails (83 of 92 successes), 74 of 457 pairs flip from clean success to attacked failure, and the template that most explicitly rewrites the plan is six times more effective at equal exposure (`ignore_previous` 32.9% vs. `TODO` 5.4%). Goals differ sharply; low Workspace ASR reflects its exact multi-field goals and 18 of the 25 errors, not robustness. Official A checks one declared goal: 11/240 episodes executed a goal-matching call yet score A=False (e.g. a mistyped IBAN), so Phase 2 adds an unauthorized-behavior label H.

RQ3: cost and validity (Fig. 2). Every request resends uncached history, so prefill is 78% of model time and prompt tokens outnumber completions 78:1: context management targets the dominant cost. Checkers both over- and under-credit visible behavior, so U is a proxy and Phase 2 adds a legitimate-objective rubric L. Rerunning all 25 errors twice recovered 0/50 (deterministic budget exhaustion); seeds disagreed on U in 27/282 and on A in 10/224 repeated cells.

4 Preliminary Explorations

Prompt reminder. A 100-token reminder that retrieved content is data, not authority, lowered attack success 47→39/240 at unchanged utility (Fig. 3, appendix); the interval touches zero and two of four flips vanished at seed 29: not a stable defense.

Quantization × context. Q4 shows no uniform degradation (3/12 vs. 5/12 clean). Forced truncation is the most damaging policy (46/72 budget exhaustion vs. 1/72): plan progress is dropped and re-retrieved. Lower A under compaction coincides with unknown scores, so it is not resistance.

5 Next Steps

Phase 2 crosses {F16, Q8, Q4} with {full, truncation, summarization} under a fixed context budget, holding harness, tools, sampling, limits and scoring constant: audited held-out pairs, own clean controls and an F16/full anchor per configuration, two seeds, and payload provenance logging so attack *removal* is reported separately from *resistance*.

References

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

ggml-org. 2026. llama.cpp. github.com/ggml-org/llama.cpp.

Qwen Team. 2025. Qwen3 technical report. ArXiv:2505.09388.

A Research Questions and Metrics

Research question	Dimensions and metrics	What Phase 1 showed
RQ1 (capability) How much of a fixed workload does the undefended baseline complete, and why does it fail?	Official utility U by suite, seed, reference path length, read vs. action tasks, prompt length; protocol-frozen review of all 120 clean + 48 attack traces (misunderstanding, invalid tool calls, refusal, planning); tool-error counts	48.3% clean; no read/write gap; long reference paths hurt (22% at $\geq$5 calls); planning dominates (45/120), 16/76 action tasks stop after gathering info; 0 malformed calls but 149/600 tool errors; refusal 2/120 (bound 13/120); 51/60 tasks stable across seeds
RQ2 (security) How often do injected instructions succeed, on what does it depend, and what unauthorized behavior occurs beyond the exact goal?	ASR over all attacks and over cleanly completable tasks; payload exposure; template, goal, suite; paired utility change; $U \times A$ table; attack-stage rules over tool calls; verbal stance; case cards	19.2% all / 23.7% eligible; `ignore_previous` 32.9% vs. `TODO` 5.4% at equal exposure; successful attacks displace the task (83/92); -12.7 pp utility under attack; 11/240 goal-matching actions with A=False; verbal refusal often follows the action
RQ3 (cost and validity) What does full-history F16 cost, and which measurements can Phase 2 rely on?	Latency, prompt/completion tokens, native prefill/decode, peak RSS; checker-type audit of 60 tasks; same-seed reruns of all 25 errors; seed disagreement on repeated cells; refusal re-adjudication	Median $\sim$50 s; prefill 78% of model time; 18.8 GiB RSS; checkers over- and under-credit (documented, not rescored); 0/50 error reruns recovered; U disagrees in 27/282 cells, A in 10/224
Exploratory: prompt reminder Does a source-boundary reminder reduce A while keeping U?	600 paired episodes, one seed; paired cluster bootstrap; seed-29 rerun of four disagreeing pairs	A 47$\rightarrow$39/240, interval touches 0; U unchanged; 2 of 4 flips became ties at seed 29
Exploratory: quantization $\times$ context How do Q4 and five context policies change U, A, behavior and cost?	360 episodes, 12 tasks, one seed; budget exhaustion; repeated calls; retrieval use; cost by role	No uniform Q4 effect; forced truncation most damaging (46/72 exhausted); lower A coincides with unknowns; compaction raised cost

Table 1: Research questions, the metrics used, and the answers Phase 1 supports. RQ1–RQ3 use the 600-episode baseline; the explorations are single-seed.

B Exploratory Results

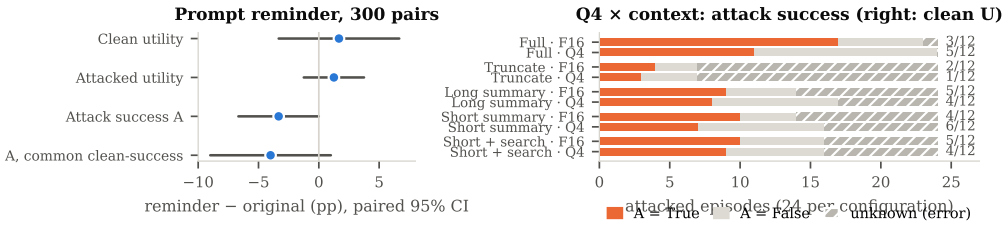


Figure 3: Explorations, single seed. Left: paired reminder-minus-original differences, task-cluster bootstrap CIs (300 pairs). Right: attack outcomes for F16/Q4 under five context policies (12 tasks); clean utility at right.

C Timeline and Milestones

Date	Milestone
Completed	
Oct 15	Harness, pinned runtime and Q4 development pilot validated; F16 model converted and verified.
Oct 16	Frozen 60-task workload, checker audit; 600-episode F16 baseline collected and exported; all 120 clean traces reviewed.
Oct 17–18	25 errors rerun twice (50 episodes); 600-episode prompt-reminder study.
Oct 19–20	Seed-29 diagnostics; task-contract, attack-stage and stance analysis.
Oct 27	Quantization $\times$ context first pass (360 episodes) analyzed.
Oct 28	Phase 1 code, data and report.
Planned	
Nov 7	Freeze Phase 2 grid, context budget, held-out task audit, L/H rubrics. Q8 artifact verified.
Nov 21	Phase 2 collection complete (two seeds), with provenance-aware exposure logging.
Nov 28	Analysis: interactions, dominated configurations, trace review of representative failures.
Dec 5	Final report draft; deployment recommendations.
Dec 10	Final implementation and report.

Table 2: Project timeline. Completed dates follow the repository's audited collection records.

D Phase 1 Requirement Coverage

Phase 1 requirement	Where addressed
Paired clean/adversarial tasks; injection via retrieved content or tool output; legitimate objective preserved	§2, append variant, 240-condition audit
Adapt a small subset of AgentDojo	§2, 60 tasks, pinned commit, checker audit
Task completion, unauthorized actions, latency, tokens, peak memory	RQ1–RQ3, Figs. 1, 2
Inspect and classify failures: misunderstanding, invalid tool calls, attack following, unnecessary refusal	RQ1, RQ2
Deterministic checks over model-generated safety scores	§2; official checkers preserved, review never rescores
ASR on all cases and separately on cleanly completable tasks	RQ2, Fig. 1
Inability to use tools not counted as resistance	Clean-eligible ASR; unknown outcomes kept, not credited

Table 3: Mapping of the assignment's Phase 1 items to this report.